$$\left.\begin{aligned} v_a &= \frac{v_0}{2} - \frac{v_0}{2} = 0 \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2} = 0 \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2} = v_0 \\ v_d &= \frac{v_0}{2} + \frac{v_0}{2} = v_0 \end{aligned}\right\} \tag{11}$$

$$\omega_1 = 2\omega_2 = \sqrt{\frac{2k_1}{m}} = \sqrt{\frac{8k_2}{m}} \tag{12}$$

$$\left.\begin{aligned} X_a &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_1}\sin\omega_1 t - \frac{l}{2} \\ X_b &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_1}\sin\omega_1 t + \frac{l}{2} \\ X_c &= \frac{1}{2}v_0 t + \frac{v_0}{\omega_1}\sin\frac{\omega_1 t}{2} + \frac{l}{2} \end{aligned}\right\} \tag{13}$$

$$X_d = \frac{1}{2}v_0 t - \frac{v_0}{\omega_1}\sin\frac{\omega_1 t}{2} + \frac{3l}{2}$$

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_1 t \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_1 t \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2}\cos\frac{\omega_1 t}{2} \\ v_d &= \frac{v_0}{2} - \frac{v_0}{2}\cos\frac{\omega_1 t}{2} \end{aligned}\right\} \tag{14}$$

$$X_b = X_c$$

$$-\frac{1}{2}\sin\omega_1 t = \sin\left(\frac{\omega_1 t}{2}\right)$$

$$\sin\omega_2 t(\cos\omega_2 t + 1) = 0 \tag{15}$$

$$\omega_2 t = n\pi \qquad n = 0, \pm 1, \pm 2, \cdots \tag{16}$$

$$\omega_2 t = \pi \tag{17}$$

$$\omega_1 t = 2\pi \tag{18}$$

$$v_b'(\quad) = -\frac{v_0}{2}$$